FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST –VII
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	D
2.	A	A	C
3.	D	A	B
4.	D	D	A
5.	C	B	D
6.	C	A	A
7.	B	C	A
8.	A	C	A
9.	A	C	B
10.	D	A	A
11.	A	A	C
12.	B	A	A
13.	A	C	B
14.	A	C	A
15.	A	A	D
16.	C	D	B
17.	C	C	C
18.	C	D	B
19.	D	B	A
1.	(A) → (q) (B) → (q) (C) → (p) (D) → (q)	(A) → (s) (B) → (q) (C) → (p) (D) → (r)	(A) → (r) (B) → (s) (C) → (p) (D) → (r)
2.	(A) → (q) (B) → (r) (C) → (p) (D) → (s)	(A) → (r) (B) → (q) (C) → (p) (D) → (q)	(A) → (q) (B) → (r) (C) → (s) (D) → (p)
3.	(A) → (q, r) (B) → (q, r) (C) → (q, r) (D) → (p, q)	(A) → (q) (B) → (p) (C) → (r) (D) → (s)	(A) → (p, r, s, t) (B) → (q) (C) → (p, q, r, t) (D) → (p)

Physics

PART – I

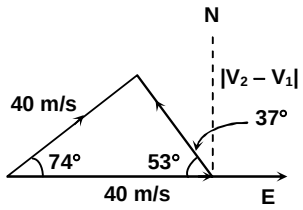
1. In equilibrium, there will not be any friction between the cylinder and the wedge. If θ be the required angle
Cylinder A: mg sin 60º = kx cos (60º – θ) (= mg cos 30º)
Cylinder B: mg sin 30º = kx cos (30º + θ) = kx sin (60º – θ)
$\therefore$ cot 30º = cot (60º – θ) $\Rightarrow$ θ = 30º.

2. $N = \frac{mv^2}{R}$
$f_{max} = mg$
$\mu N = mg$
$\mu \frac{mv^2}{R} = mg$
$v = \sqrt{\frac{Rg}{\mu}} = \sqrt{\frac{2\times 10}{0.2}}$ = 10 m/s

3. The impulse due to friction = $\mu_k \times$ impulse delivered by collision

4. The z-component of the force and the x-component of displacement are ineffective here.
$dW = F_y dy = 3xy.dy \qquad (\because z = 0)$
$= 6x^4 dx$ ($\because y = x^2$)
Integrating between x = 0 and x = 2 gives the result.

5. 144 km/hr = $144 \times \frac{5}{18}$ = 40 m/s



$$\frac{\left|\vec{V}_2 - \vec{V}_1\right|}{\sin 74^\circ} = \frac{40}{\sin(90-37^\circ)}$$

$$\Rightarrow \quad \left|V_2 - V_1\right| = \frac{40\times 2\sin 37^\circ \cos 37^\circ}{\cos 37^\circ} = 48 \text{ m/s}$$

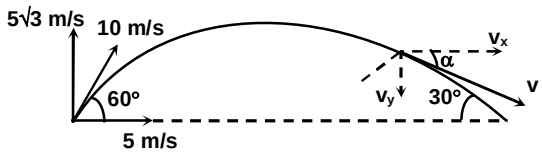
Change in momentum = $\frac{1}{3}\times$ 48 = 16

$$f_{avg} = \frac{16}{0.02} = 800 \text{ N.}$$

7. Solve the problem using μ as the coefficient of kinetic friction.

8. If $\frac{u^2\cos^2\theta}{g} > \frac{u^2\sin^2\theta}{2g}$
$\theta < \tan^{-1}\sqrt{2}$
if $\frac{u^2\cos^2\theta}{g} < \frac{u^2\sin^2\theta}{2g}$
$\theta > \tan^{-1}\sqrt{2}$.

14. Time after which velocity vector becomes perpendicular to initial velocity vector is



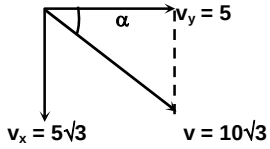
$$t = \frac{u}{g\sin\theta} = \frac{10}{10\sin 60} = \frac{2}{\sqrt{3}} \text{ seconds}$$

Let v_y be the vertical component of velocity at that instant then

$v_y = u + at$

$$v_y = 5\sqrt{3} - \frac{10\times 2}{\sqrt{3}}$$

$$v_y = -\frac{5}{\sqrt{3}}$$



$$\therefore \quad v = \sqrt{5^2 + \left(\frac{5}{\sqrt{3}}\right)^2} \Rightarrow v = \frac{10}{\sqrt{3}} \text{ m/s}$$

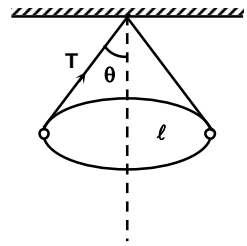
$$g\cos\alpha = \frac{v^2}{R}$$

$$R = \frac{v^2}{g\cos\alpha}$$

$$R = \frac{20}{3\sqrt{3}} \text{ m.}$$

16. $a_t = g \sin\alpha = 10 \times \dfrac{5/\sqrt{3}}{10/\sqrt{3}} = 5 \text{ m/s}^2$.

17.



19. $\omega = \sqrt{\dfrac{g}{\ell\cos\theta}}$ and $T = \dfrac{2\pi}{\omega}$

Chemistry

PART – II

SECTION – A

3. $Q = \dfrac{[C][D]}{[A][B]} = \dfrac{4 \times 5}{1 \times 2} = 10$

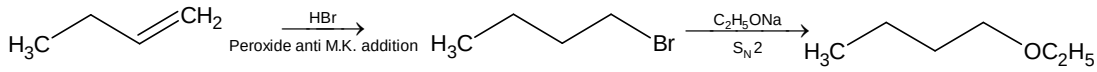
$K_c = 20$

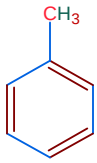
$\therefore Q < K_c$

Therefore equilibrium will shift from left to right.

4. Buffer capacity = $\dfrac{\text{moles of acid or base added per litre}}{\text{change in pH}}$

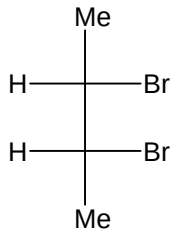
Buffer capacity = $\dfrac{\dfrac{0.2 \times 10}{1000} \times \dfrac{1000}{250}}{6.34 - 6.32} = 0.4$

5. 

6. 

will be most reactive in electrophilic substitution reaction due to hyperconjugation. It is having three α – hydrogen for hyperconjugation.

7. The product is



(Meso compound)

8. $A_{(g)} \longrightarrow 4B_{(g)} + 3C_{(g)}$

P_0	0	0	at initial
$P_0 - x$	4x	3x	at 20 minutes

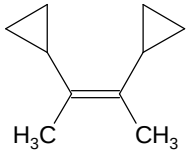
$P_0 - x + 4x + 3x = 400$

$P_0 = 100 \Rightarrow x = 50$

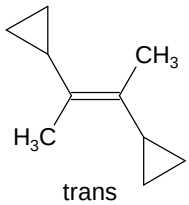
It shows 20 minutes is the half life time of the reaction.

9. $u_{RMS} = \sqrt{\dfrac{3RT}{M}} = \sqrt{\dfrac{3P}{M/V_m}} = \sqrt{\dfrac{3P}{\rho}}$

15. Hydrocarbon ($C_{10}H_{16}$)



cis



trans

SECTION – B

2. Reaction (a) is E1CB, hence RI is a carbanion.
Reaction (b) is E1, hence RI is a carbocation.
Reaction (c) is E2, hence RI is a TS.
Reaction (d) is E1, hence RI is a carbocation.

Mathematics PART – III

SECTION – A

1. All the co–ordinates of vertices are at a distance of 5 units from origin. Hence circumcentre of the triangle is (0, 0)

Centroid = $\left(\frac{3+5\cos\theta+5\sin\theta}{3}, \frac{4+5\sin\theta-5\cos\theta}{3}\right)$

Centroid divides the line joining circumcentre and orthocentre in the ratio 1 : 2

Suppose co-ordinate of orthocentre is (x_1, y_1)

$x_1 = 3 + 5\cos\theta + 5\sin\theta$ ..... (1)

$y_1 = 4 + 5\sin\theta - 5\cos\theta$ ..... (2)

Solving (1) and (2) $\sin\theta = \left(\frac{x_1+y_1-7}{10}\right) \cos\theta = \frac{x_1-y_1+1}{10}$

$(x_1 + y_1 - 7)^2 + (x_1 + y_1 + 1)^2 = 100$

Replace x_1 by x and y_1 by y

Locus of orthocentre in $(x + y - 7)^2 + (x - y + 1)^2 = 100$

2. $\frac{3}{5} = P(A) + P(B) - \frac{1}{5}$

$\Rightarrow \frac{4}{5} = 1 - P(\bar{A}) + 1 - P(\bar{B})$

$\Rightarrow P(\bar{A}) + P(\bar{B}) = \frac{6}{5}$

3. For $\arg\left(\frac{a-b}{a-c}\right) = 0 \Rightarrow \frac{a-b}{a-c} > 0$

Select 3 numbers from the set and call the greatest as 'a' and remaining two as b and c. This can be done in ${}^{2013}C_3 \times 2$ ways. Therefore the probability shall be $\frac{{}^{2013}C_3 \times 4}{{}^{2013}C_3 \times 3!} = \frac{2}{3}$

4. Put $y = -\frac{3x}{4}$ and solve for C

5. $f(x) = (2x - a)(2x - c) + (2x - b)$

So $f\left(\frac{a}{2}\right) = a - b$

$f\left(\frac{c}{2}\right) = c - b$

Now $f\left(\frac{a}{2}\right).f\left(\frac{c}{2}\right) = (a-b)(c-b) < 0$

Hence exactly one root lies between $\left(\frac{c}{2}\right)$ and $\left(\frac{a}{2}\right)$

6. Let P(x, y) be any point on the parabola.

Then by definition $\sqrt{(x-0)^2 + (y+3)^2} = y - 3 \Rightarrow x^2 = -12y$

7. $\left|\frac{z_1}{2}+\frac{z_2}{3}+\frac{z_3}{4}+\frac{z_4}{5}\right|=\frac{\lambda}{60}\left|z_1z_2z_3z_4\right|\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}+\frac{1}{z_4}\right|$

Now, $z_1\bar{z}_1=2, z_2\bar{z}_2=3, z_3\bar{z}_3=4$ and $z_4\bar{z}_4=5$ $\left(\because|z_k|=\sqrt{k+1}\right)$

$\Rightarrow \lambda=\frac{60}{\left|z_1z_2z_3z_4\right|}=\frac{60}{\sqrt{2}\sqrt{3}\sqrt{4}\sqrt{5}}=\sqrt{30}=\left|z_1z_2z_4\right|$

8. $PF_1+PF_2=15, PF_1.PF_2=52$

Let $PF_1=x, PF_2=y$

$x + y = 15$, $xy = 52$

$(F_1F_2)^2=225-104$

$(F_1F_2)^2=121\Rightarrow F_1F_2=11$

9. Let us draw tangent at P and Q which meet in T. Let $P\left(at_1^2,2at_1\right), Q\left(at_2^2,2at_2\right)$ and $R\left(at_3^2,2at_3\right)$

So $T\left(at_1t_2,\ a\left(t_1+t_2\right)\right)$

Any circle which passes through points P, Q and R must also pass through the points T as the four points are concyclic

$\Rightarrow$ Circumcentre is mid-point of TR that is M(h, k)

As normal chords at P and Q meet in R

$t_3-t_1-\frac{2}{t_1}=-t_2-\frac{2}{t_1}\Rightarrow t_1t_2=2$

$\Rightarrow h=\frac{2a+at_3^2}{2bt_1-0}=\frac{2a+a(t_1+t_2)^2}{2}$

$\Rightarrow\ t_1+t_2=\frac{2k}{a}$

From equation (1) and (2) we get $2(h-a)=a\left(\frac{2k}{a}\right)^2\Rightarrow 2y^2=a(x-a)$

10. Let P(h, 0) be a point on axis of parabola the straight line passing through P cuts the parabola at a distance r

$\Rightarrow (r\sin\theta)^2 = 4a(h + r\cos\theta)$

$\Rightarrow r^2\sin^2\theta - (4a\cos\theta)\, r - 4ah = 0$ ....(i)

Where, $r_1+r_2=\frac{4a\cos\theta}{\sin^2\theta}$

and $r_1r_2=-\frac{4ah}{\sin^2\theta}$

$\therefore\ \frac{1}{PQ^2}+\frac{1}{PR^2}=\frac{1}{r_1^2+r_2^2}=\frac{r_1^2+r_2^2}{r_1^2r_2^2}=\frac{\cos^2\theta}{h^2}+\frac{\sin^2\theta}{2ah}$

Which is constant only, if $h^2 = 2ah$ i.e. $h = 2a$

$\Rightarrow\ \frac{\cos^2\theta}{4a^2}+\frac{\sin^2\theta}{4a^2}=\frac{1}{4a^2}$

Thus, $\frac{1}{PQ^2}+\frac{1}{PR^2}=$ constant for all chords QPR, if $h = 2a$

Hence, (2a, 0) is the required point on the axis of parabola

$\therefore$ Statement-1 and statement-2 are true and statement-2 is correct explanation of statement-1

11. $f(x) = (x-3)^2(x+1)$
$\Rightarrow$ 3 is a multiple root of order 2
$f'(x) = 3x^2 - 10x + 3$
$f''(x) = 6x - 10$
$f'(3) = 0, f''(3) \neq 0$
Statement-1 is true, statement-2 is false

12. Statement-1 is correct as through $|4x^2 - 12x + 5|$
is non-differentiable at $x = \frac{1}{2}, \frac{5}{2}$ but $\cos \pi x = 0$ these points. So, $f'\left(\frac{1}{2}\right)$ and $f'\left(\frac{5}{2}\right)$ exists

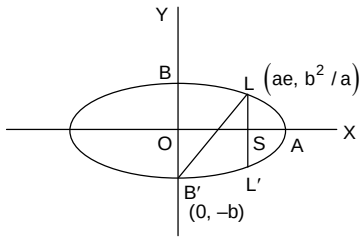
13. If each of x, y, z is less than 1/2 then, statement-1 is obviously true
Also, $1 - 2x + 1 - 2y + 1 - 2z = 3 = 2(x + y + z) = 1$
$\Rightarrow$ The sum of the three given number is positive also at must one of x, y, z can be more than 1
$\Rightarrow$ If one of x, y, z is more than or equal to $\frac{1}{2}$, then their product is less than equal to zero hence still remains true
Statement-2 is always true but it does not explain statement-1

SECTION – B

1. (A) Write the equation of normal and use the given condition

Normal at $\left(ae, \frac{b^2}{a}\right)$ is

$$\frac{x - ae}{\frac{ae}{a^2}} = \frac{y - \frac{b^2}{a}}{\frac{\left(\frac{b^2}{a}\right)}{b^2}} \Rightarrow a(x - ae) = ea\left(y - \frac{b^2}{a}\right)$$



$$\Rightarrow \quad x - ye = e\left(a - \frac{b^2}{a}\right)$$

It passes through (0, -b), $\therefore \quad b = \frac{a^2 - b^2}{a} = \frac{a^2e^2}{a} = ae^2$

$$\Rightarrow \quad b^2 = a^2e^4 \Rightarrow \quad a^2\left(1 - e^2\right) = a^2\left(e^4\right) \quad \therefore \quad e^4 + e^2 = 1$$

(C) $x^2 - 4y^2 = 1$

$$\frac{1}{4} = \left(e^2 - 1\right) \quad e = \frac{\sqrt{5}}{2}$$

(D) 1
Find the line in terms of m and satisfy the condition to solve for m
Any line through (3, 4) is $y - 4 = m(x - 3)$
According to given condition

$$\frac{m(6-3)+4-8}{\sqrt{1+m^2}} = |s| \quad \Rightarrow \quad (4m+3)^2 = 0 \quad \Rightarrow \quad m = -\frac{3}{4} = 1$$

2. We have, $P(A \cap B) = P(A) \cdot P(B) = \frac{1}{12}$

(A) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$

(B) $P(A \cup B') = 1 - (P(B) - P(A \cap B))$ = 5/6

(C) $P(B \cup A') = 3/4$

(D) $P(A \cap B) = P(A).P(B) = 1/12$

3. (A) $x\,dy = y\,dx + y^2\,dy \Rightarrow \frac{xdy - ydx}{y^2} = dy \Rightarrow -d\left(\frac{x}{y}\right) = dy$

$-\frac{x}{y} = y + c$ put x = 1, y = 1 $\Rightarrow$ c = –2

$-\frac{x}{y} = y - 2 \Rightarrow \frac{x}{3} = -5$

(B) $\frac{dy}{dt} - \frac{t}{t+1}y = \frac{1}{t+1}$

$I.f = e^{-\int \frac{t+1-1}{t+1}dt} = e^{-t+\ell n(t+1)} = (t+1)e^{-t}$

Solution is $(t + 1)e^{-t}$ $y = -e^{-t} + c$ put t = 0 and y = –1 $\Rightarrow$ c = 0

$\therefore$ $2e^{-1}\,y = -e^{-1}$

$y = -\frac{1}{2}$

(C) $(x^2 + y^2)dy = xy\,dx$

$\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$ put y = vx, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$\Rightarrow \ell nv - \frac{1}{2v^2} = -\ell nx + c$

$\therefore$ $c = -\frac{1}{2}$ $\therefore$ $\ell n\frac{y}{x} - \frac{1}{2}\frac{x^2}{y^2} = -\ell nx - \frac{1}{2}$ put y = e

$\therefore$ $x = \sqrt{3}e$

(D) $\frac{dy}{dx} + 2\frac{y}{x} = 0$

$x^2\,y = C$ put x = 1, y = 1 and we get C = 1

put x = 2 $\Rightarrow y = \frac{1}{4}$